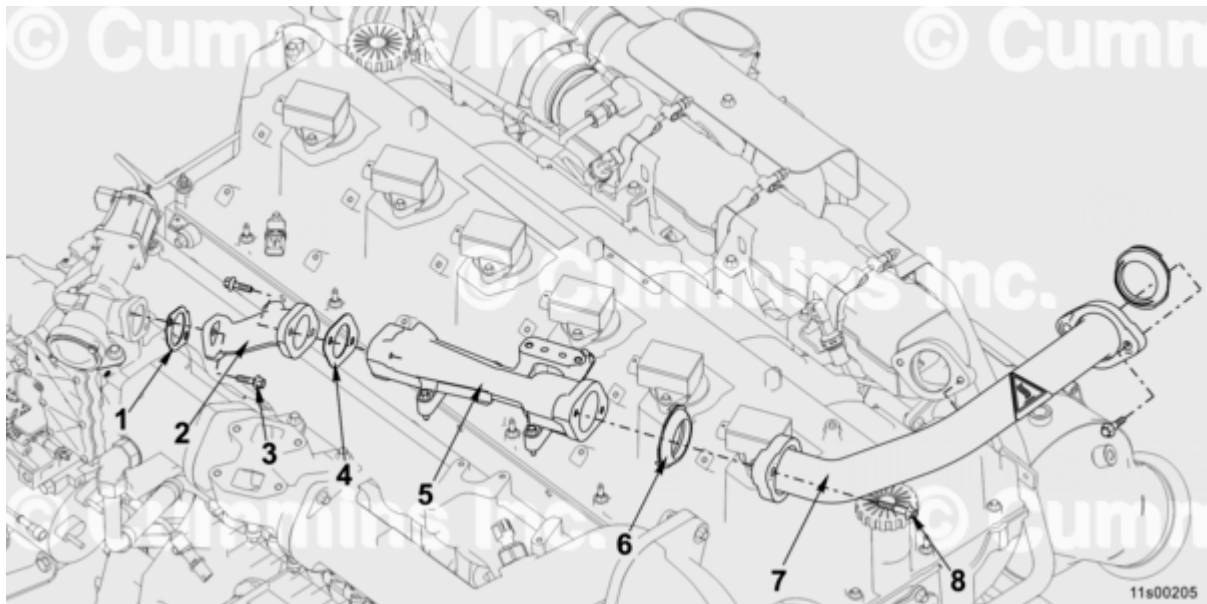


Trainings ▾

Select Service Tools



1. Connection gasket - quantity 1
2. Exhaust Gas Recirculation (EGR) crossover tube - quantity 1
3. Capscrew - quantity 4
4. Connection gasket - quantity 1
5. Venturi tube - quantity 1
6. Seal carrier - quantity 2
7. EGR crossover tube - quantity 1
8. Capscrew - quantity 4

Select Service Tools

Recommended Cummins® Service Tools

- Air Handling Clean Care Kit, Part Number 4919508

Additional Service Items

- No additional service items required.

Preparatory Steps

⚠ WARNING ⚠

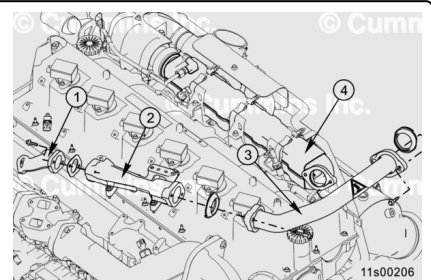
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the EGR temperature sensor. Refer to Procedure 019-378 in Section 19.
(/qs3/pubsys2/xml/en/procedures/1016/1016-019-378.html)
- Remove the EGR pressure sensor. Refer to Procedure 019-461 in Section 19.
(/qs3/pubsys2/xml/en/procedures/1016/1016-019-461.html)
- Remove the EGR differential pressure sensor. Refer to Procedure 019-370 in Section 19.
(/qs3/pubsys2/xml/en/procedures/1016/1016-019-370.html)
- Remove the EGR valve. Refer to Procedure 011-022 in Section 11.
(/qs3/pubsys2/xml/en/procedures/1016/1016-011-022.html)
- Remove the water jacket lines if equipped. Refer to Procedure 008-152 in Section 8.
(/qs3/pubsys2/xml/en/procedures/1016/1016-008-152.html)

Remove

⚠ CAUTION ⚠

Do not twist or bend the EGR connection tube during removal or installation, or damage to the hose can result.



Remove the two capscrews connecting EGR crossover tube (1) and venturi tube (2). Remove the EGR crossover tube (1).

Remove the two capscrews connecting venturi tube (2) and EGR crossover tube (3). Remove the venturi tube (2).

Remove the two capscrews connecting EGR crossover tube (3) and EGR cooler (4). Remove the EGR crossover tube (3).

Use caution to keep debris from falling into the EGR cooler and the EGR valve when removing the EGR crossover tubes.

Cover all open connections with protective caps from the Air Handling Clean Care Kit, Part Number 4919508.

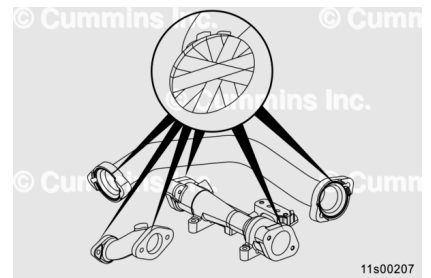
Clean and Inspect for Reuse

Scrape the gasket surfaces of EGR crossover tubes and venturi tube.

Use caution **not** to drop any foreign material into the EGR cooler, EGR valve, or venturi tube.

Note : It is common for the EGR components to have a buildup of carbon on the inside surfaces. This buildup is normal and does **not** have to be removed.

Note : If the EGR crossover tube will be off the engine for a prolonged period of time, tape over the openings in the EGR cooler and the EGR valve to prevent debris from entering the air intake system.



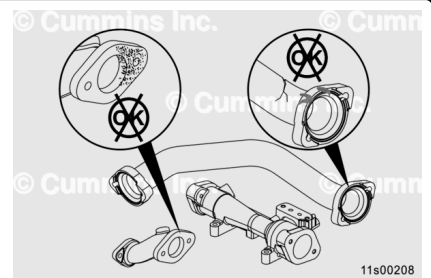
Inspect the EGR crossover tubes, carrier seals, and venturi tube for cracks, distortion, corrosion, or soot streaking on the outside of the crossover tubes and venturi tube, and indicators of an EGR leak.

Inspect for signs of coolant in the EGR crossover section tubes and venturi tube.

If signs of coolant are found in the EGR crossover tubes and venturi tube, pressure test the EGR cooler for leaks. Refer to Procedure 011-019 in Section 11.

(/qs3/pubsys2/xml/en/procedures/1016/1016-011-019.html)

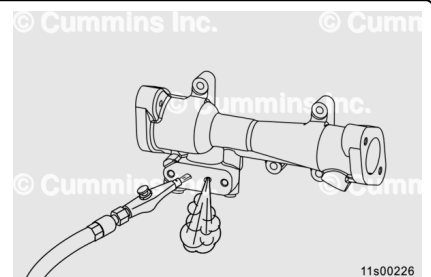
Replace the EGR crossover tubes and venturi tube if any cracks or other damage is found.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Blow shop air through the cross drillings to clear any debris or moisture.



Install

Install the EGR crossover tubes using new gaskets.

Install the four capscrews securing the venturi tube (2) to intake manifold. Hand tighten.

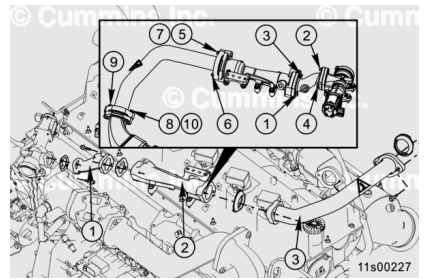
Install the capscrews connecting the EGR crossover tube (3) to the venturi tube (2) and EGR cooler. Hand tighten.

Install the capscrews connecting the EGR valve to the EGR mixer and EGR crossover tube (1). Hand tighten.

Install the capscrews connecting the EGR crossover tube (1) to EGR valve and venturi tube (2).

Tighten the capscrews connecting the EGR crossover tube (3) to EGR cooler and venturi tube (2).

Tighten the capscrews in the sequence shown.



Torque Value: 24 n•m [212 in-lb]

Tighten the four capscrews securing the venturi tube (2) to intake manifold.

Torque Value: 8 n•m [71 in-lb]

Tighten the capscrews connecting the EGR valve and the EGR mixer. Refer to Procedure 011-022 in Section 11.

(</qs3/pubsys2/xml/en/procedures/1016/1016-011-022.html>)

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the water jacket lines if removed. Refer to Procedure 008-152 in Section 8. (</qs3/pubsys2/xml/en/procedures/1016/1016-008-152.html>)
- Install the EGR differential pressure sensor. Refer to Procedure 019-370 in Section 19. (</qs3/pubsys2/xml/en/procedures/1016/1016-019-370.html>)
- Install the EGR temperature sensor. Refer to Procedure 019-378 in Section 19. (</qs3/pubsys2/xml/en/procedures/1016/1016-019-378.html>)

- Install the EGR pressure sensor. Refer to Procedure 019-461 in Section 19.
(/qs3/pubsys2/xml/en/procedures/1016/1016-019-461.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks.

Last Modified: 19-Nov-2024
